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MILITARY SPECIFICATION

DEPOSITION SYSTEMS, VACUUM, SPUTTERING TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vacuum deposition system with completely self-contained vacuum systems including all components and accessories required for thin film deposition of conductive, refractory, and dielectric material by vacuum sputtering.

1.2 Classification. The deposition systems shall be of the following types as specified (see 6.2.1).

Type I - Two electrode (diode) system. Sputtering is accomplished using an anode and cathode/target.

Type II - Three electrode (triode) system. Sputtering is accomplished using an anode, cathode, and a separate target.

Type III - Four electrode (tetrode) system. Sputtering is accomplished using an anode, cathode, target, and grid or auxiliary anode.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3433

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2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF LABOR

OSHA Safety and Health Standards
(29 CFR 1910),General Industry

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z210.1 - Metric Practice (ASTM-E-380)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

FEDERAL COMMUNICATIONS COMMISSION (FCC)

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Rules and Regulations, Vol II, Part 18, Industrial Scientific, and Medical Equipment

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete deposition system for first article inspection and approval (see 4.2 and 6.3).

* 3.2 Design. The deposition system shall be new and one of the manufacturer's current models, which meets the requirements of this specification. The deposition system shall be designed for thin film deposition of conductive, refractory, and dielectric materials under vacuum conditions, utilizing electronic sputtering processes. The deposition systems shall consist basically of a complete self-contained high vacuum system including a mechanical roughing pump, high vacuum pump, necessary baffles, and vacuum chamber; alternating current (AC), direct current (DC), and radio frequency (RF) power supplies, instrumentation, controls and enclosing cabinets. All necessary interconnecting cables and fittings shall be furnished. Non-magnetic materials shall be used where necessary to eliminate interference with operation of the deposition system. All parts subject to wear or contamination shall be accessible for service, adjustment, replacement, and repair. The design shall provide for continuous, convenient, efficient and safe operation.

* 3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and method specified in ANSI Z210.1.

* 3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as scales, stops, dial indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used, all measuring and

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indicating devices on the machine shall be graduated in the same system.

* 3.2.1.2 Dual Calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

*3.2.2 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse or sludge.

*3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

* 3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the deposition system that present safety hazards. The safety devices shall not interfere with operation of machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. Electrical interlock safety switches shall be provided for all cabinet access doors and panels (except for bolted panels), the opening of which would expose personnel to voltages of 460 volts or more. Provisions shall be made for electrically grounding the entire deposition system. All deposition parts, components, mechanism, and assemblies furnished on the unit, whether or not specifically required herein, shall comply with all of the requirements of "OSHA Safety and Health Standards (29 CFR 1910), General Industry" that are applicable to the deposition system itself Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Protective devices. The deposition systems and all components shall be provided with adequate protective devices to assure maximum and proper operation, and maximum protection against all normally expected contingencies. Unless otherwise specified (see 6.2.1), the following minimum protective devices shall be included.

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3.2.5.1 Electrical system.

a. A vacuum switch to disconnect electric power to the vacuum chamber when internal pressure exceeds a predetermined level.
b. Automatic overload protection for variable transformer, filament circuit, control, and all power supply circuits.

3.2.5.2 Vacuum system.

a. Vacuum interlocks to prevent improper sequencing of the system on systems with automatically sequencing valves.
b. Interlock to prevent operation of the high vacuum until the mechanical pump has evacuated to a preset vacuum.
c. Interlock to prevent exposure of hot diffusion pump oil to atmosphere on systems with automatically sequencing valves.
d. Thermostatically protected diffusion pump water jacket and control valves to protect diffusion pump.

* 3.2.5.3 Cooling system.

a. Means to prevent operation of the vacuum system without proper cooling of all necessary components.
b. Manual reset control to prevent automatic re-energizing of the power supply or diffusion pump in event of improper cooling.

* 3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts that require lubrication. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

* 3.2.7 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances.

* 3.3 Construction. The deposition system shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

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* 3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part. Soft solder shall not be used to attach cooling coils on diffusion pumps.

* 3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed. Where blind holes are necessary in the high vacuum environment, provision shall be made for venting gas pockets.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to preform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commericial practice.

* 3.3.5 Painting. Unless otherwise specified (see 6.2.1), the deposition system shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter detrimental to painting, and at least one coat of primer and one coat of finish are applied.

* 3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

* 3.3.7 Instruction marking and indicating plates. All words in instruction and on indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectency of the item on or to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

3.4 Components.

* 3.4.1 Vacuum pumping system. The pumping system shall include the mechanical or roughing pump, high vacuum pump, baffles, and necessary valves. The high vacuum pump may be either a diffusion pump with oil heater, a turbo

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pump or a cryopump. When only one type is acceptable it shall be as specified (see 6.2.1). The vacuum system shall be equipped with a baffle suitable for use with an external coolant as specified (see 6.2.1). All necessary devices to assure rapid pump down to the design holding and working pressure capability specified shall be included. Unless otherwise specified (see 6.2.1), the vacuum system shall be capable of evacuating the vacuum chamber filled with dry nitrogen at not less than 85 degrees F - DB (dry bulb), or not less than 85 degrees F - DB and 75 degrees F - WB (wet bulb) at atmospheric pressure to a vacuum pressure of 1 x 10-5 Torr in not more than ten minutes after cross over and shall maintain this pressure. The system shall be capable of an ultimate pressure of 1 x 10-6 Torr. All components shall be of such material, design, and fabrication to assure compatibility with the required operations and to minimize backstreaming. All components operating under high vacuum conditions shall be designed and constructed to assure that leakage into the evacuated spaces shall not exceed 1 x 10-2 Torr in one hour, starting at a pressure of 1 x 10-6 Torr with a clean, dry, empty, outgassed vacuum system and with the pumping system off. A high vacuum valve shall close off the system to allow loading of the vacuum chamber while maintaining high vacuum conditions within the pumping system. The design shall permit access to those parts requiring maintenance and cleaning. Mounting of the mechanical pump shall be such that vibration transmitted to the deposition system will be reduced to the level necessary to eliminate interference with proper operation of the system.

3.4.1.1 Valve control. Unless otherwise specified (see 6.2.1), valves in the vacuum system shall be manually controlled.

3.4.1.2 Plasma gas valves. Plasma gas fittings with control and shut-off valves shall be furnished. The control valve shall provide precise control of the gas flow to the vacuum chamber.

3.4.2 Cooling system. The cooling system shall assure that all components of the deposition systems shall be capable of continuous operation at rated capacity without exceeding design temperature conditions. Enclosures shall be ventilated as required. Forced air cooling systems shall operate satisfactorily in an ambient air temperature up to and including 96 degrees F and shall include replaceable air filters. Water cooling system shall operate satisfactorily with input water temperature of 75 degrees F to 85 degrees F and pressures of 30 to 75 PSIG. The cooling water system shall be designed to withstand 100 PSIG pressure, to allow smooth water flow, to prevent air pockets, and to facilitate drainage. The cooling water systems shall be constructed of corrosion resistant, components. Valves shall allow water flow only as required during operation. The nameplate, or

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a data plate at the input water connection, shall cite the water pressure, temperature, and flow requirements.

* 3.4.3 Vacuum chamber. Unless otherwise specified (see 6.2.1), the vacuum chamber shall be fabricated of stainless steel. The chamber dimensions, diameter and height shall be as specified (see 6.2.1). The design and construction shall be compatible with vacuum sputtering operations specified herein. When a pyrex vacuum chamber is specified, it shall be equipped with a perforated metal chamber guard. Perforations in the guard shall be such that the area inside the chamber is fully visible.

3.4.3.1 Magnetic focus coil. When specified (see 6.2.1), a magnetic coil, complete with variable voltage DC power supply, shall be furnished. The coil and power supply shall have the electrical characteristics and controls required for focusing and concentrating the plasma as necessary within the vacuum chamber.

* 3.4.4 Base plate. The vacuum chamber base plate shall be made of non-magnetic stainless steel. It may be integral with the vacuum chamber or it may be a separate unit. When furnished as a separate unit, it shall be compatible in size to the vacuum chamber specified in 3.4.3. The design and mounting of the base plate shall provide for convenient access to the underside to facilitate making connections to all feed-through connectors. In addition to the pumping port, and holes for plasma gas fittings, the base plate shall contain sufficient holes of the proper size for feed-throughs for connection of other electrical equipment, instrumentation, and mounting fixtures required for sputtering processes. All unused feed-through holes shall be vacuum sealed with removable plugs. Except for the pumping port, plugs shall be furnished for all holes in the base plate. The base plate and vacuum chamber fixtures, electrodes, and accessories shall be as specified (see 6.2.1).

* 3.4.5 Vacuum chamber hoist. Deposition systems equipped with a fixed type vacuum chamber shall have either a hinged or hoisted cover assembly for access to the chamber. Deposition systems equipped with a bell jar vacuum chamber shall be equipped with an electrically driven, hydraulically or counterbalance type operated chamber hoist or manually operated hinge assembly. The type cover and, if applicable, hoist required for fixed chamber or bell jar chamber shall be as specified (see 6.2.1). When a hoist is furnished, the hoist shall be equipped with automatic limit switch stop control at each end of travel. The hoist or hinge assembly shall permit raising bell jar type chambers to height sufficient to allow full access to the work area and to permit cleaning and removal of the chamber.

* 3.4.6 Vacuum system controls. Operator's controls on manually operated

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systems shall be conveniently located on the cabinets. All controls shall be functionally identified by permanent marking. All electrical and mechanical controls necessary for proper control of the vacuum system shall be included. Systems with automatic control shall have panel mounted controls and system condition monitoring devices. Indicating lights shall show the position of valves in the vacuum system at all time during operation.

*3.4.7 Vacuum system instrumentation. Instruments and indicator lights shall be furnished as necessary to monitor operation of the system. All monitoring instruments shall be located so as to provide good visibility to the operator located in the normal operating position. Thermocouple and ionization type vacuum gauges shall be included as required to monitor vacuum conditions. The degree of accuracy of all gauges and meters shall be as required to maintain the pressure necessary for the process involved. Special gauges required shall be as specified (see 6.2.1). Lights shall be included to indicate when power is applied to the mechanical vacuum pump, high vacuum pump, and diffusion pump oil heater if applicable. All instruments shall be complete with all necessary operating controls, and shall have the capacity to monitor operating conditions to be encountered in the vacuum system. All instruments and indicating lights shall be functionally identified by permanent marking in accordance with the manufacturer's standard commercial practice. Meters and gauges shall be calibrated directly in the units being measured and shall be permanently marked and graduated in accordance with standard commercial practice. Gauge controls shall provide for log and/or linear readouts.

* 3.4.8 Power supplies, filament and DC. Each power supply shall be furnished complete with all cables and fittings necessary for connection to the sputtering electrodes. Output voltage of all DC power supplies shall be regulated to 5 percent or better. The filament power supply shall operate from the same voltage required for operation of the vacuum system.

3.4.8.1 Power supply, AC, filament. For use with Types II and III systems. The filament supply shall be of the variable output type and adequate to power the filament at a minimum current rating of 50 amperes.

3.4.8.2 Power supply, DC, target. For use with Types I, II, and III systems. A DC power supply shall be furnished to supply power to the target. The output of the supply shall be variable through a range of at least 0 - 5000 volts with a minimum current rating sufficient to supply at least 5 milliamperes (ma) per square inch of target area. Total current capacity shall be adequate for the largest target area to be used in the vacuum chamber specified.

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3.4.8.3 Power supply, DC, anode. For use with Types II and III systems. A DC power supply shall be furnished to supply power to the anode. The output of the power supply shall be variable through a range of at least 0 - 150 volts with a minimum current rating sufficient to supply at least one ampere per square inch of anode area. Total rated current capacity shall be adequate for the largest anode area to be used in the vacuum chamber specified.

3.4.8.4 Power supply, DC, grid or auxiliary anode. For use with Type III systems. A variable output DC power supply shall be furnished to supply power to the grid or auxiliary anode used in the triode system. The output voltage shall be variable through a range of 0 - 150 volts with a rated current maximum of at least 10 amperes.

3.4.8.5 Controls and instrumentation. Each power supply covered by 3.4.8 shall be equipped with an output voltage control. The controls shall be continuously variable from zero to the maximum rated output voltage of the power supply. In addition each power supply (except filament) shall be equipped with meters to indicate the output voltage and current. The filament supply shall contain an ammeter only. All meters shall have a minimum scale length of 2.5 inches and shall be calibrated directly in amperes, milliamperes, or volts, as applicable. Voltmeters measuring voltages of 500 volts or more may be calibrated in kilovolts. All meters shall have an accuracy of $\pm$ 5 percent of full scale reading. Each power supply shall be equipped with an input power on-off switch and an indicator light to indicate when power is applied to the unit. All controls and monitoring instruments shall be mounted on a control panel located on the front of the cabinet, and shall be conveniently located and functionally identified by permanent marking in accordance with the manufacturer's standard commercial practice.

3.4.9 Power supply, radio-frequency (RF) generator. When specified (see 3.6.2), an RF generator shall be furnished for use as a power supply for RF sputtering purposes. The generator shall be furnished complete with all cables and fittings required for connection to the sputtering electrodes. Electric power shall be applied to the RF generator through conveniently located thermal overload circuit breakers with manual reset control. All necessary fuses, overload relays, and time delay circuits shall be incorporated in the generator to prevent damage to the equipment by improper operation or component failure. The generators shall meet all requirements of FCC Rules and Regulations, Volume II, Part 18, "Industrial, Scientific, and Medical (ISM) Equipment".

3.4.9.1 Power output and frequency. The rated continuous duty RF power output available at the output terminals of the generator shall be sufficient

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to supply a minimum power density to the target of 30 watts per square inch of target area. Total rated power output capacity shall be adequate for the largest target area to be used in the vacuum chamber specified. The frequency of the generator output shall be in accordance with FCC Rules and Regulations, Volum II, Part 18. Generator output power shall be variable to permit adjustment of the power input to the sputtering electrodes as required for various operating conditions.

3.4.9.2 RF generator controls. The generator final amplifier (or oscillator, when the oscillator output is applied directly to the load) shall be equipped with the following minimum controls. In addition to filament and plate power on-off switches; grid drive, plate tuning, and load controls shall be provided. In addition, an impedance matching network with necessary tuning controls shall be furnished for impedance matching between the amplifier or oscillator and the load.

3.4.9.3 Instrumentation. The following minimum instrumentation shall be furnished: DC milliameter shall be provided in the grid and plate circuits for measuring grid and plate currents. An RF wattmeter shall be provided for indicating amplifier relative power output. Indicating lights shall be installed in the amplifier filament and plate circuits to indicate when power is applied.

3.4.9.4 Cooling. Cooling the RF generator shall be accomplished as required by the use of forced air, water, or a combination of air and water. The cooling system shall meet the requirements of 3.4.2.

* 3.4.10 Cabinets. The deposition systems shall be housed in cabinets, functionally designed to facilitate operation. The frame shall be formed of steel or aluminum and constructed to support all components, except the mechanical pump which may be shock mounted or externally mounted to eliminate transfer of vibration. The side and front panels may be hinged or removable to permit access to the system for servicing. Access panels shall be equipped with quick disconnect fasteners. Adequate ventilation shall be provided for all components. Power supplies may be enclosed in individual cabinets or may be mounted in cabinet racks.

* 3.4.11 Electrical system. The deposition system shall be furnished complete with all transformers, rectifiers, timers, contactors, controls, relays, meters, suitable circuit protections, disconnect switches and all controls as required for installation and operation. Each sub-panel shall be arranged so that it may be removed independently of any other sub-panel. Solid state circuit shall be required where applicable. All terminals and units shall be clearly identifiable from the depostion systems electrical

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wiring diagram. An identified terminal for grounding the system when installed shall be mounted in or near the disconnect switch.

* 3.4.11.1 Electric service requirements. The primary electrical service required to operate the deposition system, including all power supplies, shall be as specified (see 6.2.1).

* 3.4.11.2 Motors. Motors shall be the continuous duty type and shall have sealed, permanently lubricated ball or roller bearings. Motor horsepower and speed shall provide for maximum load operating conditions.

3.5 Performance. When connected to the required electric power, plasma gas, and coolant sources, the deposition systems shall deposit thin film coatings of conductive refractory, and dielectrics, as required, under vacuum conditions. Film deposition shall be accomplished by sputtering processes. Equipped with the necessary vacuum chamber fixtures and power supplies, the system shall perform sputtering operations as required for the type specified. All sub-systems and components shall perform as specified in 3.4 and as required to perform all functions for which the system was designed. The equipment shall be capable of continous operation at rated electrical and mechanical capacity without exceeding the rated operating capacities of any components. All controls and monitoring devices shall provide for complete control and monitoring of conditions within the deposition system from no load to full load operating conditions. Meters and other instrumentation shall indicate all conditions necessary to provide for maximum efficiency under all operating conditions.

3.6 Equipment.

3.6.1 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished by the manufacturer.

One set of tools normally furnished by the manufacturer.

3.6.2 Optional equipment. When required (see 6.2.1), the following optional equipment shall be furnished as specified.

a. High capacity mechanical vacuum pump (in lieu of standard).
b. Vacuum chamber feed thru flanges.
c. Film thickness monitor
d. RF power supply.
e. Thermal overload circuit breakers with conveniently located manual reset control for installation in the power line.
f. Other options as required.

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3.7 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

* 3.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each deposition system. The nameplate shall contain the information listed below. If the deposition system is a special model, the model designation shall include the model of the basic standard deposition system and a suffix keyed to the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided that the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and rating
Coolant pressure, temperature, and volume requirements
Date of manufacture
Contract number or order number
National Stock Number or Plant Equipment Code
U S

* 3.10 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.11 Workmanship. Workmanship of the deposition system and its accessories shall be commensurate with the requirements specified herein, and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to perscribed requirements.

* 4.2 First article inspection. First article inspection shall be applied to the first article when required in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.4 and tests in 4.5. Failure of the first article to pass the examination or any first article test shall be cause for rejection.

* 4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specifed (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the test in 4.5.1, and the inspection in 4.5.4. Failure of the machine to pass the examination, test, or inspection shall be cause for rejection.

4.4 Examination. The deposition system and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.4.11 and 3.6 through 3.11.

4.5 Tests. The test requirements indicated below shall supplement, but not supersede or replace, the supplier's standard equipment test procedures. Prior to performing the following tests, each system shall be conditioned by continuous high vacuum pumping for at least 24 hours.

4.5.1 Operational test. The deposition system, including sub-assemblies shall be tested to verify conformance to the requirements of this specification.

4.5.1.1 Vacuum system test. With the vacuum system filled with dry nitrogen at not less than 85 degrees F - DB, or air at a temperature of not less than 85 degrees F - DB, and 75 degrees F - WB at atmospheric pressure, the pumping system shall be started. The pumping system shall evacuate the vacuum chamber as required in 3.4.1.

4.5.1.1.1 Vacuum system pressure rise. With a clean, dry, empty, outgassed system, at a pressure of 1 x 10-6 Torr, and with the pumping system off, the rate of pressure rise shall be measured. The pressure rise shall be measured over a period of one hour. During the one hour period, leakage into the evacuated area shall not exceed 1 x 10-2 Torr.

4.5.1.1.2 Vacuum system safety and protective equipment. All safety and protective equipment shall be tested to assure proper functioning.

4.5.1.1.3 Vacuum system, miscellaneous. All controls, timers, motors, adjusting mechanisms, vacuum sequencing systems, gauges, meters, and indi-

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cating lights shall be tested to verify proper control and monitoring of the operation of the deposition system. The diffusion pump oil heater shall be checked to determine that the oil is being maintained at the proper operating temperature. The vacuum chamber hoist, electrical system, cooling system, and gas supply system, shall be tested to assure proper operation.

4.5.1.2 Power supply tests. All controls, safety devices, and monitoring instruments on each power supply shall be tested to verify proper operation.

4.5.1.2.1 Power supplies, DC. Each power supply shall be tested with the AC line input voltage and frequency within ± 5 percent of rated value. Each power supply shall be operated for a period of at least 5 minutes under full rated load conditions. With a stable input voltage, the value of the load resistance shall be varied from no load to full rated load conditions. Output voltage regulation shall be 5 percent or better.

4.5.1.2.2 Power supply, RF (Generator) test. Each RF generator shall be tested with an AC input line voltage and frequency within ± 5 percent rated value.

4.5.1.2.2.1 RF power output. The output power shall be measured with the generator output connected, through the impedance matching network, into a load having an impedance and geometry similar to the load characteristics that will be encountered when operating as a component of a complete sputtering system. The operating voltages and currents applied to the elements of all vacuum tubes shall be within ± 5 percent of the tube manufacturer's rating for the service in use. The generator shall operated continuously for a period of at least 5 minutes at full rated power output during the test. The output power shall remain constant and shall be not less than specified (see 3.4.9.1). The generator operation shall be observed for evidence of abnormal heating of components and any evidence of unstable operation.

4.5.1.2.2.2 Operating frequency and radiation test. The manufacturer shall provide evidence, by actual testing or by certification of type approval of the generator by the FCC, that the generator meets all requirements of applicable FCC Rules and Regulations relative to the operating frequency and field strength of radiated RF energy.

4.5.1.2.2.3 Cooling system and protective devices. The operation of the cooling system and all protective and safety devices shall be tested to verify proper functioning. When water cooling is employed, the cooling system shall be checked for leaks.

4.5.1.2.2.4 Controls and instruments. All controls, meters, and other

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indicating devices shall be tested to verify proper operation.

4.5.2 Performance test. The performance capability of the complete deposition system shall be verified. The system, complete with AC, DC, and RF power supplies, as applicable, shall be connected to an appropriate electric power source, water supply for cooling, and plasma gas supply. The sputtering electrodes and vacuum chamber fixtures specified (see 3.4.4), shall be installed. A target and substrate shall be installed (see 4.5.2.1). The system shall be placed in operation and adjusted for sputtering in accordance with the manufacturer's operating manual. The sputtering process shall continue until the surface of the substrate has been completely covered by the sputtered coating. During the test, the operation of the sputtering system shall be observed for any evidence of malfunctioning. When coating of the substrate has been complete, it shall be removed from the vacuum chamber and examined. Any defects in the coating indicating deficiencies in the depostion system, or any evidence of malfunction or unstable operation of the system shall be cause for rejection.

4.5.2.1 Substrate and target. When both DC and RF power supplies are furnished, the test described in 4.5.2 shall be performed using the DC supply, and repeated using the RF supply. A target made of dielectric material shall be used when testing with the RF supply. Except as specified above, the target and substrate material and geometry shall be as determined by the manufacturer.

* 4.5.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.8.

4.5.4 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKING

* 5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended Used. This equipment is intended for use in any facility having a requirement for thin film deposition of a wide range of conducting and dielectric materials under high vacuum conditions. This equipment is designed for use in the fields of metallurgy, optics, film plating, microelectronics and integrated circuits.

* 6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their require-

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ments in procurement documents, including whether each choice is required or not required by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Type of system required (see 1.2).
c. When first article is require for inspection and approval (see 3.1 and 4.2).
d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).
e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (see 3.2.1.1).
f. Dual calibrations (US and SI), if required (see 3.2.1.2).
g. Specify additional safety and health requirements (see 3.2.4)
h. Protective devices, if different (see 3.2.5).
i. Painting, if different (see 3.3.5).
j. High vacuum pump, specify type if only one type is acceptable (see 3.4.1).
k. Type of baffle coolant, if required (see 3.4.1).
l. Vacuum pressure required, if different (see 3.4.1).
m. Automatic valve control, if required (see 3.4.1.1).
n. Vacuum chamber fabrication, if different (see 3.4.3).
o. Vacuum chamber, dimensions, diameter and height required (see 3.4.3).
p. Magnetic focus coil with power supply, if required (see 3.4.3.1).
q. Vacuum chamber fixtures, electrodes and accessories required (see 3.4.4).
r. Type of vacuum chamber cover and hoist required (see 3.4.5).
s. Special gauges, if required (see 3.4.7).
t. Primary electrical input service requirements for complete system (see 3.4.11.1).
u. Standard equipment, if different (see 3.6.1).
v. Fully describe required optional equipment (see 3.6.2).
w. Repair parts, if required (see 3.7).
x. Electromagnetic interference control, if required (see 3.8).
y. Nameplate, if different (see 3.9).
z. First article inspection, if different (see 4.2).
aa. Quality conformance inspection, if different (see 4.3).
bb. Degree of preservation, and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

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*6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. A first article comprises a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, test and approval of the first article.

* 6.4. Mercury prohibited. When the use of mercury or its compounds in the equipment and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements in this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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